THE CUMULATIVE ATTRITION RISK HYPOTHESIS

Strain, Embeddedness, and the Limits of Pay

Employees rarely leave because of one factor, and pay only helps up to a point.

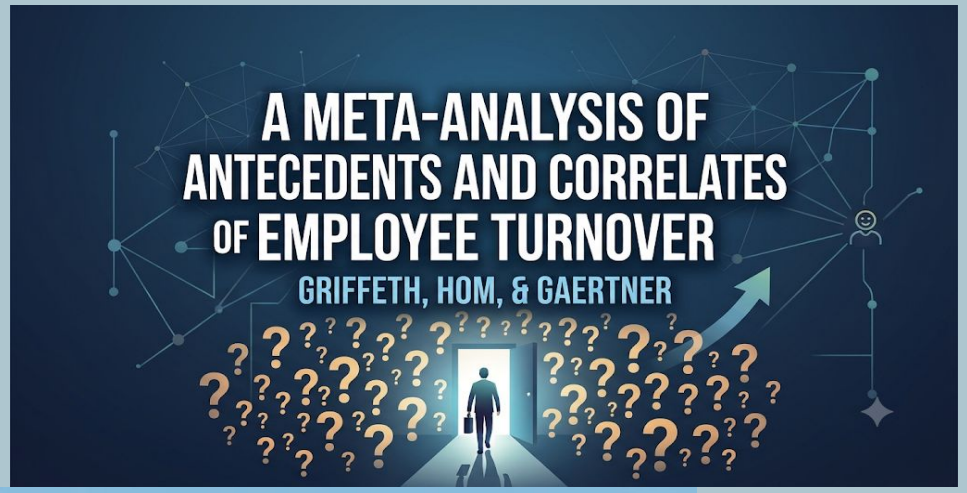
Tzlil Hayne · Oded Shmuely · Amit Shaimen · Topaz Sarid

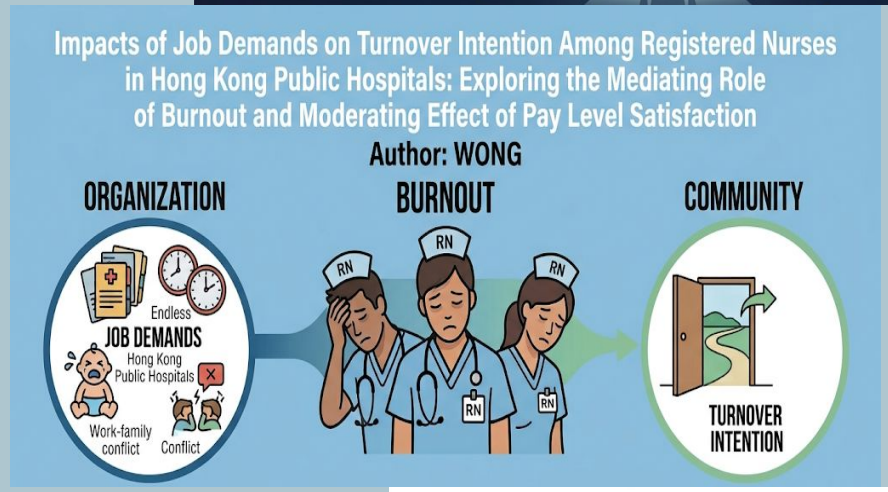
Why employee attrition matters

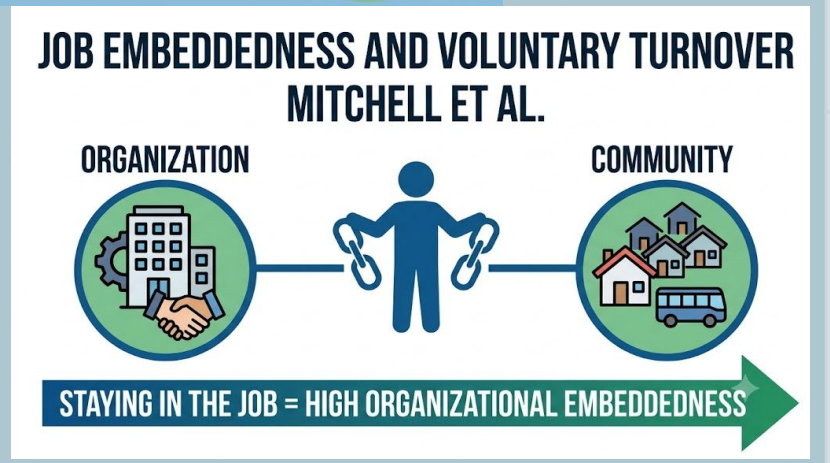
Replacing an employee is costly - and in this dataset, 16.1% of employees left.



Understanding why employees leave helps allocate resources where it matters most.







What we build on?

Prior research	Our extension
Pay predicts turnover, but is not the strongest factor (Griffeth et al., 2000)	We test whether pay still helps under high strain
Overtime and heavy demands raise work pressure and turnover (Wong et al., 2024)	We test the accumulation of factors, not each one alone
Embeddedness: workers with more ties leave less (Mitchell et al., 2001)	We use single status as a marker of lower embeddedness

RESEARCH QUESTION

How does the accumulation of overtime, business travel, and single status relate to employee attrition - and does higher pay weaken this link?

Hypothesis I:

Employee attrition increases as work-life strain markers (overtime, business travel and single status) accumulate - each additional marker is associated with higher odds of leaving.

Hypothesis II:

Higher pay is associated with lower attrition, but this association weakens as strain accumulates -pay does not fully offset the effect of high strain.

The Data

Dataset	IBM HR Analytics Employee Attrition (synthetic)
Unit of analysis	One employee per row
Employees	1,470 (no missing values)
Target	Attrition — Yes / No
Key features	Overtime, business travel, marital status, income, job level, age, Total working years

16.1%

of employees left

237 of 1,470 · imbalanced classes

How we tested it?

1

Data preparation

Clean the dataset- drop non-informative columns, and select the key variables.

2

Feature construction

Build a strain score (0-3) from overtime + travel + status.

Split income into levels: low / medium / high

3

Analysis

Fit logistic regression

Add seniority controls

Run validity checks

EmployeeCount, Over18, StandardHours, and EmployeeNumber

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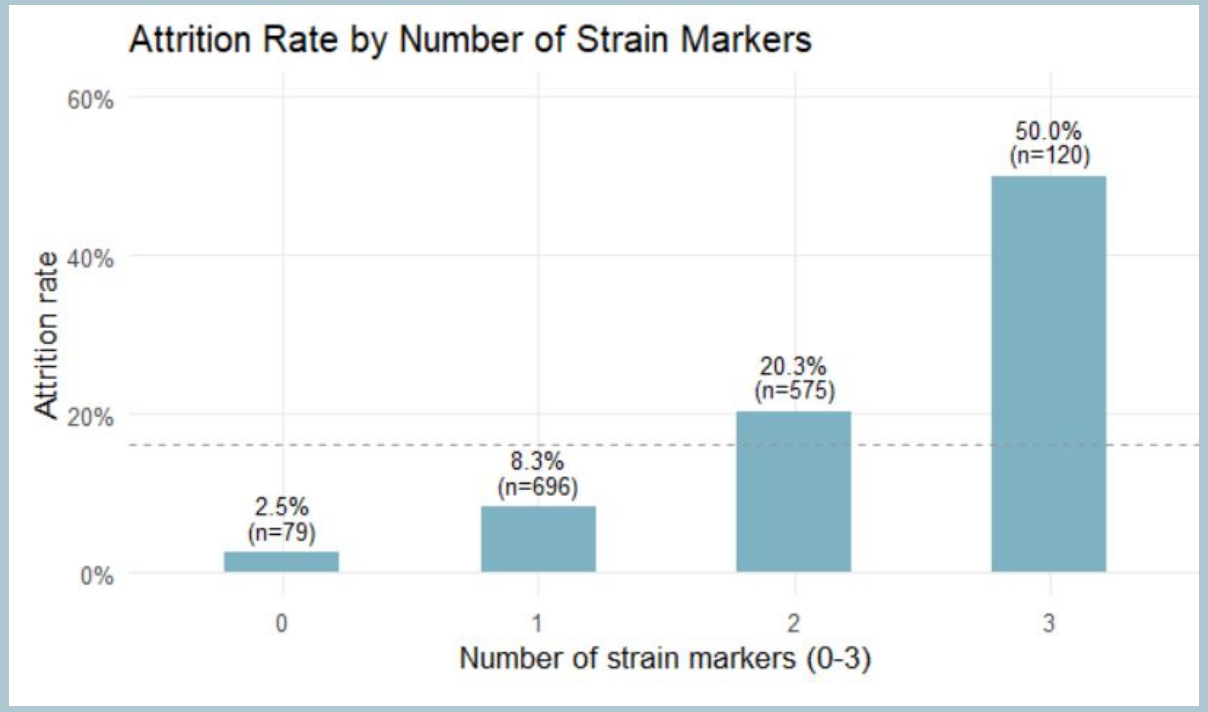
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50%

leave with all three markers

vs. just 2.5% with none - each added marker roughly triples the odds of leaving.



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Statistical Validation

Logistic regression model output for key strain metrics

Logistic regression odds ratios.

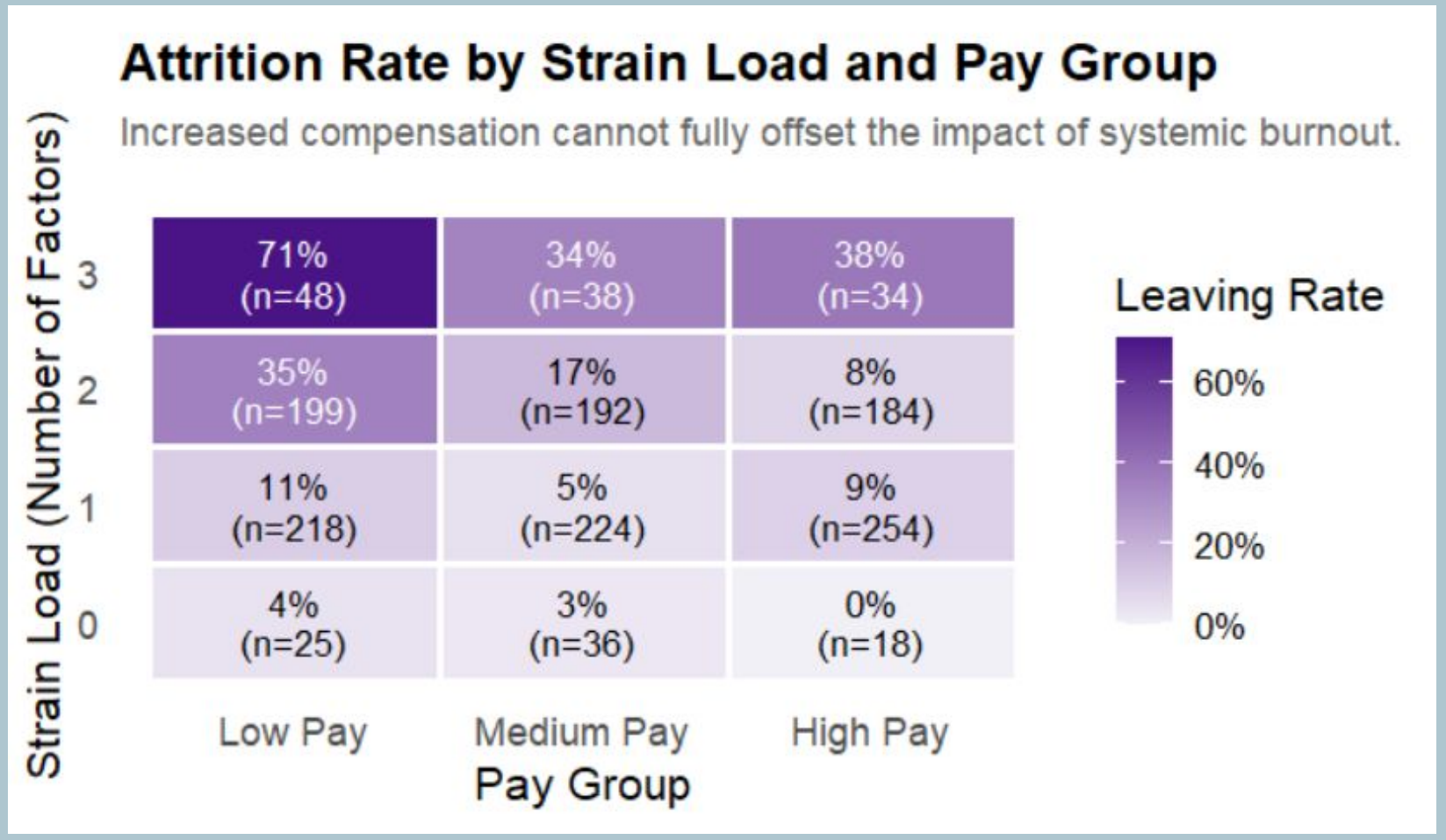
Reference groups: no overtime, non-travel, single.

Variable	OR	95% CI	p-value
Overtime: Yes	3.99	2.97–5.39	<0.001
Travel: Rarely	1.95	1.07–3.85	0.040
Travel: Frequent	3.62	1.90–7.42	<0.001
Marital: Married	0.39	0.28–0.54	<0.001
Marital: Divorced	0.29	0.19–0.45	<0.001

Key Model Insight:

Overtime, travel frequency, and marital status remain statistically significant predictors of attrition even when controlling for other variables.

Pay helps under low strain - but cannot offset high strain



At low strain, higher pay clearly lowers attrition.

At high strain, attrition stays high even for medium- and high-pay employees and extra pay no longer helps.

Validity & limitations

Validity

Controlled for age, job level.
Checked whether the income effect is really seniority.

Is the pay effect real - or just seniority?

The income OR (odds ratio - how much higher pay changes the odds of leaving) moves toward 1 and loses significance once seniority is controlled.

Model (what we control for)	Income OR (95% CI)	p-value	Significant?
Income only	0.55 (0.45-0.67)	< 0.001	Yes
+ Age	0.64 (0.52-0.80)	< 0.001	Yes
+ Job level, TotalWorkingYears	0.95 (0.57-1.58)	0.843	No

Limitations

- Dataset is synthetic
- The highest-strain pay groups are small, so exact percentages are indicative.
- the data are cross-sectional: we can identify associations between strain markers, pay, and attrition but cannot claim that strain causes employees to leave or that pay prevents leaving.

Conclusion

Strain accumulates, and pay is not a complete retention strategy.

To keep Employees , reduce stress
DON'T RELY ON PAY ONLY !

Thank you

Questions?